Electrophysiology-Based Closed Loop Optogenetic Brain Stimulation Devices: Recent Developments and Future Prospects

Paper: Kumari and Kouzani (2023 (online 2022)). Methodological review.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

What can be recorded and optically actuated, and which hardware constraints belong in a realistic closed-loop model.

How does it answer the general questions?

Signal chain. Reviews neural electrodes, acquisition electronics, feature extraction, classification, control logic and light-delivery circuitry. LFPs and action potentials provide electrical feedback; opsins translate optical stimulation into cellular effects (PDF pp. 1–7).

Usable insight. Sensor bandwidth, stimulation artifacts, processing latency, wireless power and tissue heating limit implementation. Threshold detection and PID are examples of controller approaches, not evidence that a particular controller is optimal for every circuit. A feature detector and a neural actuator should be represented as separate blocks (PDF pp. 8–15).

Model relevance. Helps specify admissible input intensity, sampling and delay, and identify observation noise/artifacts. It supplies an architecture and comparisons of existing devices, rather than one identified delayed population model. The reviewed applications have different outcomes, such as seizure detection/interruption; the review does not justify adopting beta suppression as a generic objective.

What papers does it cite that could provide more information?

- Krook-Magnuson et al. (2013), *On-demand optogenetic control of spontaneous seizures in temporal lobe epilepsy*: biological validation (ref. 98, PDF p. 18).
- Xia et al. (2017), *A low power flash-FPGA based brain implant micro-system of PID control*: embedded feedback implementation (ref. 102, PDF p. 18).